

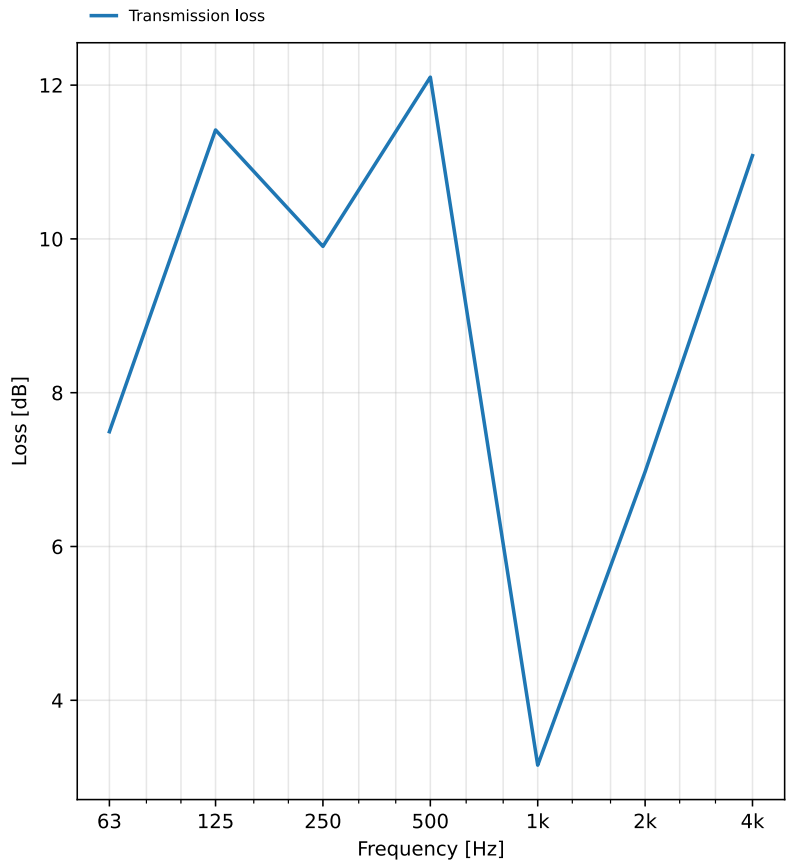
# Reactive silencer transmission loss

Predicted transmission loss of a reactive silencer, computed by the plane-wave four-pole (transfer-matrix) method (Munjal, Acoustics of Ducts and Mufflers 2nd ed., Eq. (3.27); Bies, Hansen & Howard, Engineering Noise Control 5th ed., sections 8.8 to 8.9). This is a prediction from design data, not a measurement.

|                   |                          |               |                                                       |
|-------------------|--------------------------|---------------|-------------------------------------------------------|
| Client:           | Example client           | Noise source: | Simple expansion-chamber muffler (m = 8, design case) |
| Test environment: | Duct system design study | Date of test: | 2026-07-22                                            |

Octave-band transmission loss

| f [Hz] | TL [dB] |
|--------|---------|
| 63     | 7.5     |
| 125    | 11.4    |
| 250    | 9.9     |
| 500    | 12.1    |
| 1000   | 3.2     |
| 2000   | 7.0     |
| 4000   | 11.1    |



Mean transmission loss TL = **8.9 dB**

Peak transmission loss = 12.1 dB  
Device: expansion chamber

Predicted (estimated) result computed from the declared silencer geometry by the plane-wave model without flow; it is not a measurement. Above the plane-wave cut-on frequency of the duct, and with mean flow or dissipative lining present, the realised performance departs from this model.

Result vs requirement: TL = 8.9 dB, required  $\geq 6.0$  dB → **PASS**

$TL = 10 \lg[(Z_n/Z_1) (1/4) |T_{11} + T_{12}/Z_n + Z_1 T_{21} + (Z_1/Z_n) T_{22}|^2]$  from the compound four-pole matrix T, with the characteristic impedances  $Z_1 = \rho c/S_{in}$  and  $Z_n = \rho c/S_{out}$  (Munjal Eq. (3.27), no flow). TL is the intrinsic attenuation for an anechoic termination; the insertion loss adds the source and radiation impedance mismatch.

**Laboratory:** Phonometry reference example

**Report no.:** EXAMPLE-SILENCER

**Date:** 2026-07-22

Operator: phonometry    Signature: \_\_\_\_\_

■ The results are a prediction computed from the stated inputs; they are not a measurement of a specimen.  
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